What is Javascript ?

JavaScript is a lightweight, cross-platform scripting language. It is a high-level, dynamically typed programming language used to add logic, behavior and interactivity to applications.

History of JavaScript

It was created in 1995 by Brendan Eich while he was an engineer at Netscape. It was originally going to be named LiveScript But, due to trademark reasons and certain other reasons, in December 1995, the language was finally renamed to 'JavaScript'. From then, JavaScript came into existence.

JavaScript is not a version of Java and these are two different languages with different designs, syntax details and ecosystems.

JavaScript — Decides what happens when the button is clicked
It is Dynamically Typed language - don't need to specify a variable type.
It is a Case-Sensitive language

How to Link JavaScript File in HTML ?

JavaScript can be added to HTML file in two ways:

Internal JS: We can add JavaScript directly to our HTML file by writing the code inside the <script> tag. The <script> tag can be placed inside the <body> tag.

External JS: We can write JavaScript code in another files having an extension (.js) and then link this file inside the </body> tag of the HTML file at last.

Example :-

```
<script>
        var a = 10;
        var b = 60;
        var c = a+b;
        document.write(c);
        alert("Hey There ! your answer of a + B ="+ c);
        console.log(c);
</script>
```

Difference Between Word & Keyword ?

"Word" are simply collection of characters in our real life with specific meanings, but **"Keyword"** are the system defined and specific reserved words, which hold special meaning in the programming language we are using.

word = car, book, web developer, Javascript etc.

HOW JAVASCRIPT RUN (BEHIND JS ENGINE)

1. JS CODE

You write JavaScript Code.

```
JS script.js > ...
1   let name = "Rehan";
2   const age = 30;
3   console.log(name + " & " + age);
4
5
```

This is just text (code).

2. JS ENGINE (PARSE + COMPILE)

The engine reads the code, checks syntax and converts it to machine code (so it can run).

Parse (check structure & syntax)

↓

Compile (covert to bytecode / machine code)

Now the code is ready to run.

3. OUTPUT

The engine executes the code and you see the result in the browser console

```
Rehan & 30
```

This is the final output you see.

Every Browser has its own engine. It is like an in-built software that parse the code (read line by line) & execute it.



MEMORY FLOW

JavaScript stores data in memory in two main places : **Stack** and **Heap**

Step 1 : Code is parsed

Engine finds variable and functions.

```
let name = "Rehan";
const age = 30;
```

Step 2 : Memory Allocation

Values go to **Heap** and their **references** go to **Stack**

Stack		
name	→	0xA1
age	→	0xB2

Heap		
0xA1	→	"Rehan"
0xB2	→	30

Step 3 : Execution

Wen the code runs, it reads the values from memory using those reference.

name	0xA1	→	"Rehan"
age	0xB2	→	30

INSIDE AN EXECUTION CONTEXT

Everything in Javascript happens inside the Execution Context. we can assume this execution context is like a container in which whole javasript code is executed.

After running the js code - Global Execution Context is created

Memory Component	Code Component
Variable Environment	Thread of Execution
all the variables & functions are stored in key value pair key : value a : 10 function : {....}	var name = "Rehan"; var age = 30; console.log (name + "&" + age); .....

Where does Javascript Run?

Javascript runs in a browser. These browsers (Chrome, Firefox, Safari, Edge etc;) contain Javascript engines that execute Javascript code.

Javascript can also run outside browsers using runtimes such as Node.js

What is Javascript Enging ?

A Javascript engine is like a program/softwate that reads and executes Javascript code.

What is ECMAScript?

This is one of the most important terms in Javascript.

ECMAScript is the standardized specification on which Javascript is based.

Javascript is the language people use, while ECMAScript defines the standard rules and features that Javascript implementations follow.

The standard is maintained by Ecma International through Tc39 and is specified in ECMA-262.

so, ECMAScript is Standard and Javascript is implementation/language based on that standard.

The first edition of ECMA-262 was adopted in June 1997.

It defines things such as —

– Syntax
– Data Types
– Variables
– Operators
– Functions
– Objects
– Error Handling
– Modules
– Built-in objects and functions
– Language behaviour